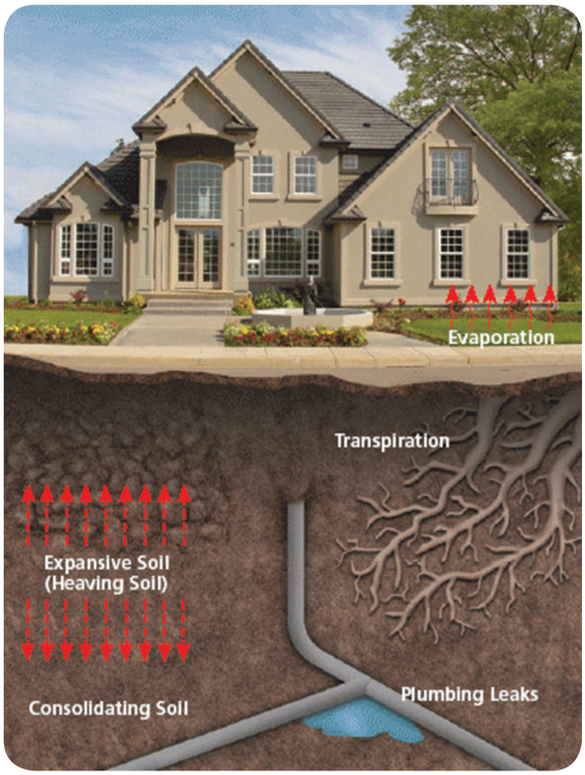
ACTIVITY 3

Collect pictures of various types of foundation, various foundation failures and prepare an album.



6.1.7.1 Unequal Settlement of the Sub Soil

When the load of all the parts of building is not even, the unequal settlement occurs. Based on the intensity of the load it may be low or high. This is because the bearing capacity of soil which is not uniform in all places. Cracks are formed in the building due to the variations in the settlement of soil.

Method of Prevention

i. Foundations should rest on hard rock or hard moorum.

ii. Type and design of foundation should be selected according to the nature of soil by considering the safe bearing capacity of the soil.

DO YOU KNOW?

The tallest building in the world.

'Jeddah Tower' is a skyscraper under construction in Jeddah, Saudi Arabia. If completed in 2020 as planned, the Jeddah Tower will reach an unprecedented height, becoming the tallest building in the world as well as the first structure to reach the one-kilometer-high mark.



6.1.7.2 Unequal Settlement of the Masonry

The mortar joints in the wall and other building portion may shrink and this may lead to unequal settlement of the building portion.

Method of Prevention

i. The water used to mix the cement mortar should be in correct proportion.

ii. Height of raising the super structure should be uniform. It should not exceed 1.5m / day.

iii. Curing of the masonry should be adequate.

6.1.7.3 Withdrawal of Moisture from the Sub Soil

Failure of foundation occurs where there is variation in the height of water table. The cracks are formed when the soil shrink due to the sudden reduction of water table from top to bottom.

Method of Prevention

The foundations are provided with piles to the extreme or should rest on hard rock in those places.

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